

Comparative effects of salt substitutes on blood pressure, cardiovascular events and mortality: a systematic review and network meta-analysis

Received: 17 Aug 2025

Accepted: 07 Jan 2026

Published online: 23 January 2026

Honghao Lai, Gihad Nesrallah, Gordon Guyatt, Robin Vernooij, Jiayi Liu, Weilong Zhao, Liangying Hou, Bei Pan, Jiajie Huang, Norm Campbell, Jinhui Tian, Kehu Yang, Long Ge & Bradley Johnston

Cite this article as: Lai, H., Nesrallah, G., Guyatt, G. *et al.* Comparative effects of salt substitutes on blood pressure, cardiovascular events and mortality: a systematic review and network meta-analysis. *BMC Med* (2026). <https://doi.org/10.1186/s12916-026-04635-z>

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

Comparative effects of salt substitutes on blood pressure, cardiovascular events and mortality: a systematic review and network meta-analysis

Honghao Lai, *doctoral student*^{1,2}, Gihad Nesrallah, *nephrologist, assistant professor*^{3,4}, Gordon H. Guyatt, *methodologist, distinguished professor*⁵, Robin W.M. Vernooij, *assistant professor*^{6,7}, Jiayi Liu, *masters student*^{1,2}, Weilong Zhao, *masters student*^{1,2}, Liangying Hou, *doctoral student*^{5,8}, Bei Pan, *methodologist, doctoral student*⁸, Jiajie Huang, *masters student*⁹, Norm R.C. Campbell, *professor*¹⁰, Jinhui Tian, *methodologist, professor*^{6,11}, Kehu Yang, *methodologist, professor*^{6,11}, Long Ge*, *methodologist, professor*^{1,2,11}, Bradley C. Johnston*, *methodologist, professor*^{12,13}

1. Department of Health Policy and Management, School of Public Health, Lanzhou University, Lanzhou, China
2. Evidence-Based Social Science Research Center, School of Public Health, Lanzhou University, Lanzhou, China
3. Division of Nephrology, Humber River Hospital, Toronto, Ontario, Canada
4. Department of Medicine, Temerty Faculty of Medicine, Toronto, Ontario, Canada
5. Department of Health Research Methods, Evidence, and Impact, McMaster University, Hamilton, Ontario, Canada
6. Department of Nephrology and Hypertension, University Medical Center Utrecht, Utrecht, The Netherlands
7. Julius Center for Health Sciences and Primary Care, University Medical Center Utrecht, Utrecht University, Utrecht, The Netherlands
8. Evidence-Based Medicine Center, School of Basic Medical Sciences, Lanzhou University, Lanzhou, China
9. College of Nursing, Gansu University of Chinese Medicine, Lanzhou, China
10. Department of Medicine, Libin Cardiovascular Institute, University of Calgary, Calgary, Alberta T3L1S7, Canada
11. Key Laboratory of Evidence Based Medicine and Knowledge Translation of Gansu Province, Lanzhou, China
12. Department of Nutrition, College of Agriculture and Life Sciences, Texas A&M University, College Station, TX, USA
13. Department of Epidemiology and Biostatistics, School of Public Health, College Station, TX, USA

*Corresponding author:

Long Ge, Evidence-Based Social Science Research Center, School of Public Health, Lanzhou University, No. 199 Donggang West Road, Chengguan District, Lanzhou 730000; Email, gelong2009@163.com. (ORCID 0000-0002-3555-1107)

Bradley C. Johnston, Department of Nutrition, College of Agriculture and Life Sciences, Texas A&M University, College Station, TX, USA; bradj49@gmail.com

Financial support: This study was funded by the Fundamental Research Funds for Central Universities of Lanzhou University (lzujbky-2024-oy11).

Conflicts of interests: None.

Abstract**Background**

Hypertension is a major risk factor for cardiovascular disease. Salt substitutes may reduce sodium intake while maintaining palatability, but comparative effects across formulations remain uncertain.

Methods

We conducted a systematic review and frequentist random-effects network meta-analysis of randomised controlled trials in adults comparing salt substitutes with regular salt, other substitutes, or no intervention. Databases (PubMed, Embase, CENTRAL, CNKI, Wanfang), WHO-ICTRP and ClinicalTrials.gov were searched from inception to Oct 3, 2025 (PROSPERO CRD42023451859). We assessed the risk of bias using a modified Cochrane tool and conducted a random-effects network meta-analysis, with evidence certainty evaluated through the GRADE approach.

Results

We included 34 randomised controlled trials involving 37,063 participants across 15 countries (17 from China, 17 from other countries; mean age 62.3 years). Our results indicate that moderate-potassium and low-sodium salt substitutes (25–40% KCl, 60–79% NaCl) probably reduce all-cause mortality, cardiovascular mortality, non-fatal cardiovascular events, and systolic blood pressure (SBP) compared to regular salt, with reductions of 7–17 deaths per 1,000 individuals and 4.39–4.64 mmHg for SBP, based on moderate to high certainty evidence. Mortality and cardiovascular benefits are predominantly driven by one large Chinese trial (SSaSS, n=20,995); excluding this trial eliminated statistical significance for all-cause mortality. Among non-Chinese studies, none contributed mortality data. Substitutes with higher potassium or very low sodium showed similar blood pressure reductions but provided less certain evidence regarding mortality and events. No substitute increased adverse events or withdrawals, and acceptability was comparable to regular salt.

Conclusions

Salt substitutes, particularly moderate-potassium and low-sodium formulations, represent a promising sodium reduction strategy. However, current evidence for

mortality and cardiovascular event benefits is dominated by one large Chinese trial and has very limited generalisability beyond Chinese populations with high discretionary salt use. These products appear acceptable and safe in people without renal impairment, but clinicians should rule out kidney disease and hyperkalaemia risk before recommending them, and large trials in non-Chinese populations are needed.

Keywords: salt substitutes, cardiovascular risk, hypertension, mortality

ARTICLE IN PRESS

Background

Increased systolic blood pressure (SBP), a modifiable cause of cardiovascular disease (CVD) accounts for 16.0% of deaths globally with hypertension consuming up to 8.5% of GDP in industrialized nations.[1–3] Dietary intake of sodium chloride (NaCl) is a key factor in the onset of hypertension,[4,5] prompting recommendations for salt reduction to reduce CVD risk and mortality.[6] However, challenges arise in reducing population-level salt intake due to individual preferences and palatability issues, as well as broader supply-level factors in food production, distribution, and marketing.[7]

Salt substitutes alter salts' composition while preserving its flavour, and represent a promising intervention for reducing sodium intake.[8] Salt substitutes are products that replace NaCl with potassium chloride (KCl) or other constituents such as magnesium salts or chitosan.[9] Recent systematic reviews have established the effectiveness of salt substitutes in reducing blood pressure and cardiovascular events.[10] However, variability in salt substitute formulations may lead to variations in their efficacy and acceptability.[11] The comparative effectiveness of different substitute types and compositions remains unclear, limiting evidence-based selection.

We therefore conducted this first-ever systematic review and network meta-analysis, applying GRADE and minimal important difference (MID) thresholds, to compare the effects of widely available salt substitutes on blood pressure, non-fatal cardiovascular events, and mortality, and to assess acceptability and safety.

Methods

Data sources and searches

We prepared this manuscript in accordance with the PRISMA-NMA guidelines,[12] and adhered to a registered protocol (PROSPERO–CRD 42023451859; Additional file 1: Text S1). Pairs of reviewers independently performed the study selection, data

extraction and risk-of-bias assessments with discrepancies resolved through discussion or with adjudication from a third author, if required.

We searched PubMed, Embase, Cochrane Central Register of Controlled Trials, China National Knowledge Infrastructure, Wanfang database, the World Health Organization's International Clinical Trials Registry Platform and ClinicalTrials.gov from inception to October 3, 2025. The search strategy (Additional file 1: Table S1) employed a combination of controlled vocabulary and keywords pertaining to salt substitutes, blood pressure, cardiovascular diseases, and randomised controlled trials (RCTs). We also reviewed reference lists of included studies and relevant reviews.

Study selection

We included RCTs that enrolled adults (≥ 18 years) comparing salt substitutes (substituting a minimum of 3% of NaCl with KCl or other constituents), with either regular salt ($\geq 97\%$ NaCl), other salt substitutes or no intervention. We excluded trials involving combined interventions (e.g., salt substitutes and salt restriction). Important outcomes included all-cause mortality, cardiovascular mortality, non-fatal cardiovascular events, and systolic and diastolic blood pressure (DBP). Other outcomes included withdrawals due to adverse events (AEs), acceptance, adverse events related to intervention, withdrawals due to any reason, serum potassium, serum sodium, urinary potassium, and urinary sodium.

Data extraction and quality assessment

We used Rayyan,[13] a web-based platform to assist with study selection. We screened titles and abstracts using selection criteria detailed in Additional file 1: Table S2, then evaluated potentially eligible full-texts. We extracted study characteristics, demographic information, description of interventions and comparators, and outcomes (Additional file 1: Table S3). For crossover RCTs, we included only the first period. We assessed the risk of bias using a modified Cochrane 1.0 instrument.[14] For cluster-RCTs, we followed the Cochrane handbook (Additional file 1: Text S2).[15]

Data synthesis and analysis

Given the complexity of salt substitute formulations and uncertainty about which compositional features drive health outcomes, we classified salt substitutes using three

strategies (Table 1): (1) by the presence of specific components comprising at least 3% of the salt mixture, assuming that variations in composition may lead to different health outcomes;[16] (2) by KCl content, categorized as high (50 to 65%), moderate (25 to 40%) and low (5 to 20%) potassium salt substitutes, assuming that potassium levels affect health outcomes;[11] and (3) by NaCl content, categorized as reduced (80 to 97%), low (60 to 79%), very low (40 to 59%) and extremely low (2 to 39%) sodium salt substitutes, assuming that sodium levels affect health outcomes.[17]

We computed continuous variables as the mean difference (MD) using change scores when reported by at least 50% of studies, and otherwise we used the final (endpoint) values.[18] For dichotomous outcomes, we calculated risk ratios (RRs) along with 95% confidence intervals (CIs). We used available Cochrane guidance to impute missing values (Additional file 1: Text S3),[19] and to extract graphically-represented data when necessary.

Assuming similar heterogeneity across all comparisons and considering the correlations induced by multi-arm studies, we conducted random-effects arm-based network meta-analyses using a frequentist weighted least squares methodology.[20] For direct comparisons, we executed DerSimonian-Laird random-effects pairwise meta-analyses. We evaluated statistical heterogeneity within each pairwise and network meta-analysis comparison using I^2 and Q . [21] We calculated indirect estimates from the network through node splitting and back calculation methods (i.e., subtracting direct from network estimates). For every enclosed loop within the network, we examined local inconsistency using both back calculation and separate assessment of indirect and direct evidence.[22] We evaluated the transitivity assumption by comparing the distributions of key population and intervention study characteristics among studies grouped by comparison.[23] Where at least ten studies were available, we assessed small-study effects and publication bias using funnel plots.[24]

We conducted network meta-regression analyses to explore the impact of mean age, proportion of females, mean body mass index (BMI), baseline SBP, and baseline DBP (Additional file 1: Text S4). We used the ICEMAN instrument for assessing the

credibility of effect modification analyses.[25] We also conducted sensitivity and additional analyses including exclusion of trials conducted in Chinese populations, trials in hypertensive patients, and subgroup analyses by intervention duration for blood pressure outcomes.

GRADE certainty assessment and summary of evidence

For each network meta-analysis, we evaluated the certainty of evidence based on estimates of the minimally important difference (MID) threshold for direct, indirect, and network comparisons.[26,27] We initially assigned high certainty to direct comparisons, rating down for risk of bias, inconsistency, indirectness, and publication bias. For indirect evidence, we rated certainty based on the lowest certainty among direct comparisons in the most dominant first order loop, further rating down for intransitivity.[28,29] In our evaluation of network evidence, we began with the certainty derived from the dominant direct and/or indirect comparisons and further considered incoherence or imprecision.[30] We used the MID thresholds for mortality, non-fatal cardiovascular events, and blood pressure, based on estimates used in previous dietary guidelines and systematic reviews.[31–33] For fatal outcomes, the threshold was 5 events per 1,000; for non-fatal outcomes the threshold was 10 events over five-years;[31,32] while for SBP and DBP (mean of follow-up 10.5 months), the thresholds were 3.0 and 2.0 mmHg, respectively.[34] For other continuous outcomes (serum and urinary electrolytes), we specified the MID as 0.3 times the baseline standard deviation (SD);[35] for other dichotomous outcomes, we specified the MID as documented in Additional file 1: Table S4. We rated down for imprecision if the 95% confidence intervals (CIs) of pooled estimates crossed our MID thresholds.[36,37] For dichotomous outcomes, we calculated anticipated absolute effects by applying the summary RRs or ORs to baseline risks (Additional file 1: Text S3 outlines the formulas).[38] We defined intermediate baseline risk as a 5-10% five-year cardiovascular event risk and high baseline risk as a 20-30% five-year cardiovascular event risk.[32,39] The baseline risks were ascertained using control group data from a meta-analysis conducted by the Cholesterol Treatment Trialists' Collaboration (Additional file 1: Text S5).[39]

Role of the funding source

There was no funding source for this study.

Results

Our search yielded 11,424 records with 39 additional records identified through reference lists. After screening, 34 studies including 37,063 participants from 15 countries were deemed eligible (Figure 1).[40–73] A detailed list of excluded studies with reasons for exclusion is provided in Additional file 1: Table S2.

Table 2 and Additional file 2: Table S1 summarise the characteristics of the included studies, encompassing 34 studies with a weighted mean age of 62.3 years, a weighted mean proportion of females of 46.5%, a weighted mean BMI of 25.5, a weighted mean SBP of 147.6 mmHg, a weighted mean DBP of 85.8 mmHg, and a weighted mean prevalence of hypertension of 69.1%, with 40.5% using antihypertensives. Salt quantities were comparable across groups and trials. Additional file 2: Table S2 presents the recruitment criteria of trials; 16 trials explicitly excluded individuals with renal impairment. The included studies were conducted in various countries, with half (17/34 studies) being carried out in China. The remaining studies were performed in Brazil (2), France (2), Italy (2), Korea (1), Peru (1), South Africa (1), Netherlands (1), UK (1), Japan (1), Iran (1), USA (1), Norway (1), Finland (1), and India (1). Among the trials not conducted in China, none examined mortality outcomes, while one trial in Peru showed statistically significant and clinically important effects on non-fatal cardiovascular events, reducing the events by 44 and 148 per 1,000 in intermediate and high cardiovascular risk populations, respectively.[45]

The results of risk of bias assessments detailed in Additional file 2: Table S3 reveal that of 21 parallel and 3 crossover trials, 13 (54.2%) had a low risk of bias, while among ten cluster trials, seven (70.0%) had a low bias. The trials assessed at high risk of bias were rated down most often because of missing outcome data (7/11, 63.6%),

lack of reported blinding (5/11, 45.5%), and selective outcome reporting (4/11, 36.4%).

Table 1 presents the three classifications of salt substitutes and their corresponding formulations. Based on primary composition, salt substitutes were categorized into five types: K-salt, K-Ca-salt, K-Mg-salt, K-Ca-Mg-salt, and Chitosan salt (Figure 2a). Among these, K-salt, composed of varying proportions of NaCl and KCl, has the most substantial evidence base (16 trials, 12,510 participants). K-salts showed important reductions in all-cause mortality, cardiovascular mortality, non-fatal cardiovascular events, and SBP among individuals at intermediate and high baseline risk, without increasing the risk of adverse events or withdrawals compared to regular salt, based on low to high certainty evidence (Table 3a; Additional file 3). To further investigate the impact of salt substitute composition on health outcomes, we conducted NMAs using two classification approaches: one based on KCl content and another on NaCl content (Figures 2b and 2c; Additional files 4 and 5).

Mortality Outcomes

Seven trials involving 26,037 participants reporting all-cause mortality and three trials with 24,384 participants reporting cardiovascular mortality showed that substitutes with moderate-potassium containing 25 to 40% KCl (6 trials; 24,106 participants) and low-sodium containing 60 to 79% NaCl (5 trials; 23,959 participants) probably reduced mortality over five years compared to regular salt (Tables 3b and 3c), based on moderate to high certainty evidence.

For all-cause mortality, both formulations (i.e., moderate-potassium and low-sodium) reduced deaths by seven per 1000 followed, among individuals at intermediate cardiovascular risk over 5 years, based on moderate certainty evidence. For those at high cardiovascular risk, we found 17 fewer deaths per 1,000 over 5 years, based on high certainty evidence.

For cardiovascular mortality, both substitutes reduced deaths by five per 1,000 at intermediate risk and by 16 per 1,000 at high risk over a 5-year period, with both findings supported by moderate certainty evidence.

The mortality benefits of moderate-potassium and low-sodium salt substitutes were primarily supported by a large Chinese RCT (n=20,995 participants).[61] After excluding this study, while the significant all-cause mortality benefit was eliminated (still showing reductions of 4 and 10 deaths/1000 for those at intermediate and high cardiovascular risk), cardiovascular mortality remained significantly reduced, with 9 and 29 fewer deaths per 1,000 in intermediate and high risk populations, respectively, over 5 years. Additionally, the certainty in effects for high-potassium and very low-sodium (e.g., 50% NaCl + 50% KCl) salt substitutes were less pronounced with only very low to low certainty evidence for mortality outcomes. No significant difference was found between the substitutes.

Non-fatal cardiovascular events

Ten trials with 26,505 participants reported the risk of non-fatal cardiovascular events, including angina (n=2),[41,62] stroke (n=1),[63] major adverse cardiovascular events (n=1),[61] and unspecified or unclear events (n=6).[45,48,51,65,69,73] Nine of ten trials were included in the analysis while one was excluded because there were no events. Moderate-potassium and low-sodium (9 trials; 26,048 participants) substitutes were probably beneficial for intermediate risk individuals and might be beneficial for high cardiovascular risk individuals compared to regular salt, based on low to moderate certainty evidence over five years. However, these effects were also influenced by the same large Chinese RCT,[61] and excluding it made the benefits insignificant (while low-sodium substitutes still showing reductions of 24 and 80 deaths/1000 in intermediate and high risk populations, respectively; Additional file 6: Table S1). The certainty in effects for high-potassium, low-potassium, reduced-sodium, and very low-sodium salt substitutes were less pronounced and very low to low (Table 3b). No significant difference was found between the substitutes. After excluding six studies with unspecified cardiovascular events, the results did not change.

Blood pressure

Twenty-nine trials with 27,631 participants reported blood pressure, of which 28 trials encompassing 27,621 participants with available data were included in the

analysis. High-potassium (e.g. 50% NaCl + 50% KCl) and moderate-potassium (e.g. 62.5% NaCl + 25% KCl) salt substitutes probably reduced SBP compared to regular salt, based on moderate certainty evidence. Reduced-sodium (e.g. 87% NaCl + 6.5% KCl) and very low-sodium (e.g. 50% NaCl + 50% KCl) salt substitutes reduced both SBP and DBP compared to regular salt, based on moderate to high certainty evidence. Low-potassium (e.g. 87% NaCl + 6.5% KCl), low-sodium salt substitutes, and extremely low-sodium (e.g. 33% NaCl + 67% KCl) salt substitutes showed potential effects on blood pressure reduction, though these findings were supported by low to very low certainty evidence, and salt substitutes with lower NaCl content did not demonstrate enhanced effects, possibly due to limited sample size resulting in less precise measurement.

The results of meta-regression indicated that salt substitutes may have a larger effect on both SBP and DBP in older patients, and a larger effect on SBP in those with higher baseline SBP. The credibility of these interaction effects assessed using ICEMAN subgroup criteria was judged as low (Additional files 4 and 5). One trial that provided conclusions only (no explicit data) aligned with the current findings (Additional file 6: Table S2).[42] No significant difference was found in the comparisons between salt substitutes on SBP. Intervention duration did not significantly affect blood pressure reduction (Additional file 6: Figure S1).

Adverse events related to intervention and withdrawals due to adverse events

Among ten trials comprising 7,060 participants, a total of 117 AEs related to intervention were reported, averaging 2 per 100 person-years of follow-up. However, the network meta-analysis only included five trials with 3,336 participants and 117 AEs, excluding five trials due to zero-cells. With moderate certainty evidence, high-potassium, moderate-potassium, reduced-sodium, low-sodium, and very low-sodium salt substitutes demonstrated trivial to no difference in AEs compared to regular salt. The evidence regarding AEs for low-potassium and extremely low-sodium salt substitutes was less conclusive and rated as low certainty. No significant difference was found between substitutes. Qualitative descriptions of AEs across the trials corroborated these findings, indicating that the salt substitutes did not

significantly increase the risk of AEs, even among elderly individuals with chronic kidney disease or those taking potassium-sparing medications.

Ten trials involving 3,311 participants documented a total of 10 withdrawals due to AEs, with an incidence rate of 3 per 1000 person-years of follow-up. However, a quantitative analysis on this outcome was not conducted due to zero-cells across most trials. Further examination of these reported cases suggested that the AEs leading to the withdrawals were likely unrelated to the salt substitutes.

Other outcomes

Six trials, involving 1,881 participants, reported data on acceptance by conducting surveys on acceptability of salt substitutes and recording withdrawals due to non-acceptance. No evidence suggested a poorer acceptance of the salt substitutes compared to regular salt.

Twenty-four trials comprising 26,152 participants reported 4,636 withdrawals, of which 19 trials comprising 25,584 participants were included in the analysis. Compared to regular salt, moderate-potassium salt substitutes probably have trivial or no withdrawal risk over five years based on moderate certainty evidence, while the certainty in effects for other substitutes were less pronounced and only very low to low.

Regarding electrolyte balance, very low to moderate certainty evidence suggested that increased KCl content, coupled with correspondingly reduced NaCl levels, may lead to elevated urinary potassium excretion and decreased urinary sodium excretion.

Discussion

This systematic review and network meta-analysis of 34 RCTs involving 37,063 participants from 15 countries provides a comprehensive evaluation of salt substitutes with varied compositions. In patients primarily with hypertension and without kidney disease at intermediate to high cardiovascular risk, our NMA of salt substitutes suggests that K-salts, composed of varying proportions of NaCl and KCl, demonstrate important reductions in all-cause mortality, cardiovascular mortality, non-fatal

cardiovascular events and SBP. While recent mechanistic study suggests that potassium enrichment may contribute more substantially to blood pressure lowering effects than sodium reduction,[74] our NMA of salt substitutes with different KCl and NaCl content did not detect significant differences in blood pressure reduction between formulations, and further investigation is required to clarify these mechanisms across diverse populations and settings. Compared to regular salt, moderate-potassium and low-sodium salt substitutes (i.e., 25 to 40% KCl and 60 to 79% NaCl) demonstrate the most compelling evidence, with probable reductions in all-cause mortality (7 to 17 fewer deaths/1,000 followed at intermediate and high cardiovascular risk, respectively) and cardiovascular mortality (5 to 16 fewer deaths/1,000 followed at intermediate and high cardiovascular risk, respectively) over 5 years, based on moderate to high certainty evidence. These substitutes may also reduce the risk of non-fatal cardiovascular events and decrease SBP, albeit with low to moderate certainty evidence. Notably, these substitutes appear to maintain favourable safety and acceptability profiles, with no significant increase in AEs or palatability concerns. Salt substitutes with higher or lower KCl and NaCl content (e.g., 50% NaCl + 50% KCl) demonstrate significant and important effects on blood pressure, but the certainty of evidence for mortality and cardiovascular events is very low to low due to limited study population size. Although higher potassium concentrations in salt substitutes may theoretically confer greater benefits,[75] our analysis shows that while these formulations demonstrate larger point estimates in SBP reduction, the difference between various concentrations based on our MID estimates is not important.

Multiple previous reviews concluded that the blood pressure-lowering effects of salt substitutes are generalizable across diverse populations and geographic regions,[8,76,77] with mortality benefits similarly extrapolated to global settings.[8] With a larger evidence base, our findings reveal limitations overlooked by previous reviews and confirm Greenwood et al.'s cautious interpretation that current evidence,[78] dominated by a single large Chinese trial (SSaSS, n=20,995 participants), has very limited generalizability beyond populations following Chinese dietary patterns characterized by high discretionary salt use and home food

preparation. Our sensitivity analysis reinforces this concern: excluding the SSaSS trial eliminated the statistically significant all-cause mortality benefit, demonstrating that pooled mortality effects are driven predominantly by this single geographic and cultural context. Beyond confirming this geographic limitation with a larger dataset (34 vs 16 trials), our NMA provides three novel contributions: first, formulation-specific guidance through head-to-head comparisons revealing that moderate-potassium substitutes have more robust mortality evidence than high-potassium substitutes; second, systematic application of minimal important difference thresholds demonstrating that moderate-potassium formulations exceed clinically meaningful thresholds; and third, transparent GRADE certainty ratings for each formulation, enabling evidence-based selection of specific salt substitute types rather than treating them as a homogeneous intervention class.

The concentration of evidence in Chinese populations, who tend to prepare food at home more often than many Western populations, have higher sodium intake levels, and consume less processed or ultra-processed foods,[79–81] may limit the global generalizability of our findings. Although point estimates continued to show reductions in mortality after excluding the SSaSS trial (4 and 24 fewer deaths/1,000 in intermediate risk populations for all-cause mortality and non-fatal cardiovascular events, respectively), the loss of statistical significance underscores the need for large-scale trials in other populations.

Further research is needed to establish effectiveness and safety in diverse populations with varied food systems,[82] including non-Chinese populations where the proportion of food prepared outside the home is higher,[80,81] as well as Black populations that are more likely to have sodium sensitivity.[83,84] The uncertainty surrounding the effectiveness and safety of other compositions warrants further investigation. Caution should be exercised when considering increased potassium fortification of the food supply, as this may pose a risk of hyperkalemia and sudden cardiac death among those with impaired renal function and other risk factors.[85]

When considering the implementation of salt substitutes, it is crucial to account for palatability from a geographical and cultural perspective related dietary sodium intake.

The acceptability of reduced-sodium products together with products with increased potassium may vary across populations, particularly between China and Western countries, potentially influencing the effectiveness of low sodium substitutes.[79,86,87] Moreover, while salt substitutes did not appear to significantly increase the risk of AEs or withdrawals due to AEs compared to regular salt in the overall population, safety concerns persist for patients with chronic or undiagnosed kidney disease or renal impairment. In addition, if patients are on potassium sparing diuretics, they may be more susceptible to hyperkalaemia and at risk of sudden death due to impaired potassium excretion.[88] Still further, patients with type 1 or 2 diabetes may also be at risk of hyperkalemia.[89–91] Hyperkalemia can cause cardiac arrhythmias and, in severe cases, sudden cardiac death, with evidence suggesting its association with cardiac events and cardiovascular mortality across all levels of kidney function, even in individuals without prior kidney problems.[92,93] Clinicians should rule out kidney problems before suggesting salt substitutes and future studies should exclude or possibly prioritize the evaluation of safety and efficacy in some of these vulnerable populations (e.g., diabetes). Although our study did not find significant concerns regarding the acceptability of salt substitutes, evidence suggests that some individuals may not tolerate the taste of foods with high potassium content, potentially leading to reduced adherence to low-sodium salt substitutes.[94,95] There is however evidence that some patients can increase that tolerance and acceptability of salt substitutes.[96]

With respect to study strengths, this is the first systematic review with network meta-analysis to report on the cardiovascular health outcomes of salt substitutes with varied compositions and different KCl and NaCl content, expanding upon prior meta-analyses which relied entirely on pairwise analyses only.[76,77] Our network meta-analysis has allowed for a comparative effectiveness analysis of 7 salt substitutes (combining different potassium and sodium content classifications), determining that moderate-potassium and low-sodium salt substitutes (i.e. 25 to 40% KCl and 60 to 79% NaCl) has the best available evidence for patient important mortality and major cardiovascular events. Compared to prior reviews, we also

assessed acceptability, safety, evidence certainty using GRADE methods, and we determined the comparative effectiveness of available salt substitutes using MID thresholds, all of which are important factors to optimally support clinical decision-making and guideline development.

Our study is not without limitations. The predominance of Chinese participants may limit the generalizability of the findings to other populations. Additionally, variations in taste perception and adherence across different salt substitute formulations might have influenced the results. Our analysis could not fully account for total dietary sodium and potassium intake beyond the substitutes, and the inconsistent relationship between reduced NaCl content in salt substitutes and urinary sodium excretion suggests that incomplete replacement of regular salt or additional dietary sodium sources may have confounded the results. The safety assessment in special populations, such as those with chronic kidney disease or on potassium-sparing medications, was insufficient, necessitating targeted future investigation. Moreover, consistent with a recent systematic review,[10] the long-term effectiveness and safety of salt substitutes remains to be established. The validation of urinary sodium and potassium excretion as markers of adherence and effectiveness also requires further research. Lastly, the current categorization of potassium salt dosages was simplified for the purpose of the network meta-analysis; future studies could explore more refined network models using original dosage data, while cautioning against overfitting.

Conclusions

This systematic review and network meta-analysis provides evidence that salt substitutes, particularly moderate-potassium and low-sodium formulations, probably reduce all-cause and cardiovascular mortality, as well as SBP, making them, in addition to WHO suggestions such as choosing minimally processed low sodium food products,[97] a promising sodium reduction strategy with the caveat that individuals without kidney impairment at intermediate and high cardiovascular risk need more

investigation. However, the evidence supporting these mortality benefits is derived predominantly from a single large Chinese trial (SSaSS, n=20,995). Sensitivity analysis excluding this trial eliminated statistical significance for all-cause mortality, demonstrating very limited generalizability beyond Chinese populations with high discretionary salt use and home food preparation. Among non-Chinese trials, none examined mortality outcomes. The applicability to other populations, including other Asian populations with different dietary patterns or higher processed food consumption, remains highly uncertain. These substitutes demonstrated favorable safety and acceptability profiles in patients without renal impairment. Clinicians should first rule out underlying kidney problems and hyperkalemia risk before recommending salt substitutes.

ARTICLE IN PRESS

List of Abbreviations

AE: Adverse event

BMI: Body mass index

CI: Confidence interval

CKD: Chronic kidney disease

CVD: Cardiovascular disease

DBP: Diastolic blood pressure

GDP: Gross domestic product

GRADE: Grading of Recommendations Assessment, Development and Evaluation

ICEMAN: Instrument to assess the Credibility of Effect Modification Analyses

KCl: Potassium chloride

MACE: Major adverse cardiovascular events

MD: Mean difference

MID: Minimal important difference

NaCl: Sodium chloride

NMA: Network meta-analysis

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

RCT: Randomised controlled trial

RR: Risk ratio

SBP: Systolic blood pressure

SD: Standard deviation

SSaSS: Salt Substitute and Stroke Study

WHO: World Health Organization

WHO-ICTRP: World Health Organization International Clinical Trials Registry Platform

Additional Files Declaration

Additional file 1: Text S1. Modifications to the Protocol. Table S1. Full search strategy. Table S2. Study eligibility. Table S3. Data collection. Text S2. Risk of bias assessments. Text S3. Processing of data. Text S4. Priori hypotheses of network meta-regression analyses. Table S4. MID thresholds of each outcome. Text S5. Certainty of evidence assessments.

Additional file 2: Table S1. Baseline characteristics of included studies. Table S2. Participants inclusion and exclusion criteria of included studies. Table S3. Results of risk of bias assessments.

Additional file 3: Figure S1. Network plots for the outcomes (including network plots for all-cause mortality, cardiovascular mortality, non-fatal cardiovascular events, systolic and diastolic blood pressure, adverse events related to intervention, withdrawals due to any reason, serum potassium and sodium concentrations, and 24-hour urinary potassium and sodium excretion). Figure S2. League tables (including absolute estimates for all-cause mortality and cardiovascular mortality at intermediate and high baseline risk, absolute estimates for non-fatal cardiovascular events at intermediate and high baseline risk, estimates for systolic and diastolic blood pressure, absolute estimates for withdrawals due to any reason and adverse events related to intervention, estimates for serum potassium and sodium, and estimates for urinary potassium and sodium). Figure S3. Forest plots for network evidence (including forest plots of network meta-analysis estimates for all-cause mortality, cardiovascular mortality, non-fatal cardiovascular events, systolic and diastolic blood pressure, adverse events related to intervention, withdrawals due to any reason, and serum potassium concentration).

Additional file 4: Figure S1. Network plots for the outcomes (including network plots for all-cause mortality, cardiovascular mortality, non-fatal cardiovascular events, systolic and diastolic blood pressure, adverse events related to intervention, withdrawals due to any reason, serum potassium and sodium concentrations, and 24-hour urinary potassium and sodium excretion). Figure S2. League tables (including

absolute estimates for all-cause mortality and cardiovascular mortality at intermediate and high baseline risk, absolute estimates for non-fatal cardiovascular events at intermediate and high baseline risk, estimates for systolic and diastolic blood pressure, absolute estimates for withdrawals due to any reason and adverse events related to intervention, estimates for serum potassium and sodium, and estimates for urinary potassium and sodium). Figure S3. Forest plots for network evidence (including forest plots of network meta-analysis estimates for all-cause mortality, cardiovascular mortality, non-fatal cardiovascular events, systolic and diastolic blood pressure, adverse events related to intervention, withdrawals due to any reason, and serum potassium concentration).

Additional file 5: Figure S1. Network plots for the outcomes (including network plots for all-cause mortality, cardiovascular mortality, non-fatal cardiovascular events, systolic and diastolic blood pressure, adverse events related to intervention, withdrawals due to any reason, serum potassium and sodium concentrations, and 24-hour urinary potassium and sodium excretion). Figure S2. League tables (including absolute estimates for all-cause mortality and cardiovascular mortality at intermediate and high baseline risk, absolute estimates for non-fatal cardiovascular events at intermediate and high baseline risk, estimates for systolic and diastolic blood pressure, absolute estimates for withdrawals due to any reason and adverse events related to intervention, estimates for serum potassium and sodium, and estimates for urinary potassium and sodium). Figure S3. Forest plots for network evidence (including forest plots of network meta-analysis estimates for all-cause mortality, cardiovascular mortality, non-fatal cardiovascular events, systolic and diastolic blood pressure, adverse events related to intervention, withdrawals due to any reason, and serum potassium concentration).

Additional file 6: Table S1. Network meta-analysis of salt substitutes after exclusion of the Neal, 2021 trial. Table S2. Outcomes not amenable to meta-analysis. Figure S1. Additional analysis for effects of salt substitutes on blood pressure reduction at different intervention times.

Declarations

Acknowledgments: Not applicable.

Consent for Publication: All authors agreed to the publication of this manuscript.

Funding: This study was funded by the Fundamental Research Funds for Central Universities of Lanzhou University (lzujbky-2024-oy11). The funder had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Data Availability: All data generated or analysed during this study are included in this published article and its supplementary information files. The datasets supporting the conclusions of this article are available in the additional files.

Authors' Contributions: LG and HHL conceived the study idea. HHL, LG, BCJ, and GN designed the analysis. LG and HHL designed and performed the search strategy. HHL, JJH, BP, and JYL screened abstracts and full texts, extracted data, and judged the risk of bias of included studies. HHL performed the data analysis. HHL wrote the first draft of the manuscript. LG, BCJ, GN, RV, NRCC, and GG provided important content and methodological support on early drafts. All authors interpreted the data analysis and critically revised the manuscript. LG, BCJ, GN, BP, LYH, WLZ, RV, BCJ, and GG provided suggestions, technical support or material support. LG, BCJ, GN, JHT, and KHY supervised the study. HHL is the guarantor. The corresponding author attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted. All authors read and approved the final manuscript.

Competing interests: LG, GG, RV and BCJ are members of the GRADE Working Group. While unrelated to this work, BCJ received a grant from Texas A&M AgriLife Research to fund investigator-initiated research related to saturated and polyunsaturated fats. The grant was from Texas A&M AgriLife institutional funds from interest and investment earnings, not a sponsoring organization, industry, or company. BCJ also holds NIDDK R25 funding to support the development of evidence-based nutrition educational resources, resources that draw on GRADE Working Group methods. Remaining authors declare no conflicts of interest.

Ethics Approval and Consent to Participate: This systematic review and network meta-analysis used publicly available published data from previously conducted studies. Ethical approval was not required for this secondary analysis of published data, as per institutional guidelines and the Declaration of Helsinki. All included studies obtained appropriate ethical approval and participant consent as reported in their original publications.

ARTICLE IN PRESS

Legends

Table 1 Classification and definitions of salt substitutes

The same formulation may be classified differently under each approach. For example, 75% NaCl + 25% KCl would be classified as K-salt (Strategy 1), Moderate-potassium (Strategy 2), and Low-sodium (Strategy 3).

KCl: potassium chloride; NaCl: sodium chloride; MgSO₄: magnesium sulfate; K: potassium; Ca: calcium; Mg: magnesium.

Table 2 List of included studies and baseline characteristics

SBP: systolic blood pressure; DBP: diastolic blood pressure; IG: intervention group; BMI: body mass index; NR: not report; NaCl: sodium chloride; KCl: potassium chloride; MgSO₄: magnesium sulfate.

Table 3 Summary of network meta-analysis results

SS: Salt substitutes; SBP: Systolic blood pressure; DBP: Diastolic blood pressure.

Certainty of evidence ratings were assessed for the populations actually studied in the included trials, without pre-specification of particular target populations. When applying this evidence to populations with substantially different dietary patterns, food preparation methods, baseline sodium intake, or genetic backgrounds compared to the study populations, practitioners may need to consider downgrading evidence certainty for indirectness.

Figure 1 Literature screening flow diagram

Figure 2 Network plots of all included studies

References

- 1 Wierzejska E, Giernaś B, Lipiak A, *et al.* A global perspective on the costs of hypertension: A systematic review. *Arch Med Sci.* 2020;16:1078–91. doi: 10.5114/aoms.2020.92689
- 2 Ettehad D, Emdin CA, Kiran A, *et al.* Blood pressure lowering for prevention of cardiovascular disease and death: A systematic review and meta-analysis. *The Lancet.* 2016;387:957–67. doi: 10.1016/S0140-6736(15)01225-8
- 3 GBD compare | institute for health metrics and evaluation. <https://www.healthdata.org/data-tools-practices/interactive-visuals/gbd-compare> (accessed 9 October 2025)
- 4 Messerli FH, Hofstetter L, Syrogiannouli L, *et al.* Sodium intake, life expectancy, and all-cause mortality. *Eur Heart J.* 2021;42:2103–12. doi: 10.1093/eurheartj/ehaa947
- 5 Huang L, Trieu K, Yoshimura S, *et al.* Effect of dose and duration of reduction in dietary sodium on blood pressure levels: Systematic review and meta-analysis of randomised trials. *BMJ.* 2020;368:m315. doi: 10.1136/bmj.m315
- 6 Santos JA, Tekle D, Rosewarne E, *et al.* A systematic review of salt reduction initiatives around the world: A midterm evaluation of progress towards the 2025 global

- non-communicable diseases salt reduction target. *Adv Nutr Bethesda Md.* 2021;12:1768–80. doi: 10.1093/advances/nmab008
- 7 Nurmilah S, Cahyana Y, Utama GL, *et al.* Strategies to reduce salt content and its effect on food characteristics and acceptance: A review. *Foods Basel Switz.* 2022;11:3120. doi: 10.3390/foods11193120
 - 8 Yin X, Rodgers A, Perkovic A, *et al.* Effects of salt substitutes on clinical outcomes: A systematic review and meta-analysis. *Heart Br Card Soc.* 2022;108:1608–15. doi: 10.1136/heartjnl-2022-321332
 - 9 Cepanec K, Vugrinec S, Cvetković T, *et al.* Potassium chloride-based salt substitutes: A critical review with a focus on the patent literature. *Compr Rev Food Sci Food Saf.* 2017;16:881–94. doi: 10.1111/1541-4337.12291
 - 10 Greenwood H, Barnes K, Clark J, *et al.* Long-term effect of salt substitution for cardiovascular outcomes: A systematic review and meta-analysis. *Ann Intern Med.* 2024;177:643–55. doi: 10.7326/M23-2626
 - 11 Greer RC, Marklund M, Anderson CAM, *et al.* Potassium-enriched salt substitutes as a means to lower blood pressure: Benefits and risks. *Hypertension.* 2020;75:266–74. doi: 10.1161/HYPERTENSIONAHA.119.13241
 - 12 Hutton B, Salanti G, Caldwell DM, *et al.* The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: Checklist and explanations. *Ann Intern Med.* 2015;162:777–84. doi: 10.7326/M14-2385
 - 13 Ouzzani M, Hammady H, Fedorowicz Z, *et al.* Rayyan—a web and mobile app for systematic reviews. *Syst Rev.* 2016;5:210. doi: 10.1186/s13643-016-0384-4
 - 14 Methods commentary: Risk of bias in randomized trials 1. DistillerSR. <https://www.distillersr.com/resources/methodological-resources/risk-of-bias-commentary> (accessed 9 October 2025)
 - 15 Higgins JP, Savović J, Page MJ, *et al.* Assessing risk of bias in a randomized trial. *Cochrane Handbook for Systematic Reviews of Interventions.* John Wiley & Sons, Ltd 2019:205–28.
 - 16 Xu X, Zeng L, Jha V, *et al.* Potassium-enriched salt substitutes: A review of recommendations in clinical management guidelines. *Hypertension.* 2024;81:400–14. doi: 10.1161/HYPERTENSIONAHA.123.21343
 - 17 Mente A, O'Donnell M, Yusuf S. Sodium intake and health: What should we recommend based on the current evidence? *Nutrients.* 2021;13:3232. doi: 10.3390/nu13093232
 - 18 Higgins JP, Green S, editors. *Cochrane handbook for systematic reviews of interventions: Cochrane book series.* 1st edn. Wiley 2008.

- 19 Pearson MJ, Smart NA. Reported methods for handling missing change standard deviations in meta-analyses of exercise therapy interventions in patients with heart failure: A systematic review. *PLoS ONE*. 2018;13:e0205952. doi: 10.1371/journal.pone.0205952
- 20 Balduzzi S, Rücker G, Nikolakopoulou A, *et al.* **netmeta**: An R package for network meta-analysis using frequentist methods. *J Stat Softw*. 2023;106. doi: 10.18637/jss.v106.i02
- 21 Rücker G. Network meta-analysis, electrical networks and graph theory. *Res Synth Methods*. 2012;3:312–24. doi: 10.1002/jrsm.1058
- 22 Dias S, Welton NJ, Caldwell DM, *et al.* Checking consistency in mixed treatment comparison meta-analysis. *Stat Med*. 2010;29:932–44. doi: 10.1002/sim.3767
- 23 Cipriani A, Higgins JPT, Geddes JR, *et al.* Conceptual and technical challenges in network meta-analysis. *Ann Intern Med*. 2013;159:130–7. doi: 10.7326/0003-4819-159-2-201307160-00008
- 24 Chaimani A, Salanti G. Using network meta-analysis to evaluate the existence of small-study effects in a network of interventions. *Res Synth Methods*. 2012;3:161–76. doi: 10.1002/jrsm.57
- 25 Schandelmaier S, Briel M, Varadhan R, *et al.* Development of the instrument to assess the credibility of effect modification analyses (ICEMAN) in randomized controlled trials and meta-analyses. *Can Med Assoc J*. 2020;192:E901–6. doi: 10.1503/cmaj.200077
- 26 Hultcrantz M, Rind D, Akl EA, *et al.* The GRADE working group clarifies the construct of certainty of evidence. *J Clin Epidemiol*. 2017;87:4–13. doi: 10.1016/j.jclinepi.2017.05.006
- 27 Zeng L, Brignardello-Petersen R, Hultcrantz M, *et al.* GRADE guidelines 32: GRADE offers guidance on choosing targets of GRADE certainty of evidence ratings. *J Clin Epidemiol*. 2021;137:163–75. doi: 10.1016/j.jclinepi.2021.03.026
- 28 Brignardello-Petersen R, Tomlinson G, Florez I, *et al.* Grading of recommendations assessment, development, and evaluation concept article 5: Addressing intransitivity in a network meta-analysis. *J Clin Epidemiol*. 2023;160:151–9. doi: 10.1016/j.jclinepi.2023.06.010
- 29 Zeng L, Brignardello-Petersen R, Hultcrantz M, *et al.* GRADE guidance 34: Update on rating imprecision using a minimally contextualized approach. *J Clin Epidemiol*. 2022;150:216–24. doi: 10.1016/j.jclinepi.2022.07.014
- 30 Izcovich A, Chu DK, Mustafa RA, *et al.* A guide and pragmatic considerations for applying GRADE to network meta-analysis. *BMJ*. 2023;e074495. doi: 10.1136/bmj-2022-074495
- 31 Johnston BC, Zeraatkar D, Han MA, *et al.* Unprocessed red meat and processed meat consumption: Dietary guideline recommendations from the nutritional recommendations (NutriRECS) consortium. *Ann Intern Med*. 2019;171:756–64. doi: 10.7326/M19-1621

- 32 Karam G, Agarwal A, Sadeghirad B, *et al.* Comparison of seven popular structured dietary programmes and risk of mortality and major cardiovascular events in patients at increased cardiovascular risk: Systematic review and network meta-analysis. *BMJ*. 2023;e072003. doi: 10.1136/bmj-2022-072003
- 33 Ge L, Sadeghirad B, Ball GDC, *et al.* Comparison of dietary macronutrient patterns of 14 popular named dietary programmes for weight and cardiovascular risk factor reduction in adults: Systematic review and network meta-analysis of randomised trials. *BMJ*. 2020;m696. doi: 10.1136/bmj.m696
- 34 Sedaghat AR. Understanding the minimal clinically important difference (MCID) of patient-reported outcome measures. *Otolaryngol Neck Surg*. 2019;161:551–60. doi: 10.1177/0194599819852604
- 35 Woaye-Hune P, Hardouin J-B, Lehur P-A, *et al.* Practical issues encountered while determining minimal clinically important difference in patient-reported outcomes. *Health Qual Life Outcomes*. 2020;18:156. doi: 10.1186/s12955-020-01398-w
- 36 Brignardello-Petersen R, Guyatt GH, Mustafa RA, *et al.* GRADE guidelines 33: Addressing imprecision in a network meta-analysis. *J Clin Epidemiol*. 2021;139:49–56. doi: 10.1016/j.jclinepi.2021.07.011
- 37 Wang Y, Devji T, Carrasco-Labra A, *et al.* A step-by-step approach for selecting an optimal minimal important difference. *BMJ*. 2023;e073822. doi: 10.1136/bmj-2022-073822
- 38 Skoetz N, Goldkuhle M, Van Dalen EC, *et al.* GRADE guidelines 27: How to calculate absolute effects for time-to-event outcomes in summary of findings tables and evidence profiles. *J Clin Epidemiol*. 2020;118:124–31. doi: 10.1016/j.jclinepi.2019.10.015
- 39 Cholesterol Treatment Trialists' (Ctt) Collaborators. The effects of lowering LDL cholesterol with statin therapy in people at low risk of vascular disease: Meta-analysis of individual data from 27 randomised trials. *The Lancet*. 2012;380:581–90. doi: 10.1016/S0140-6736(12)60367-5
- 40 Allaert FA. Double-blind, randomized, crossover, controlled clinical trial of NaCl + chitosan 3% versus NaCl on mild or moderate high blood pressure during the diet and lifestyle improvement period before possible prescription of an antihypertensive treatment. *Int Angiol J Int Union Angiol*. 2013;32:94–101.
- 41 Allaert F-A. Effect of NaCl + chitosan 3% vs. NaCl on high blood pressure parameters of healthy volunteers with prehypertension. *Minerva Cardioangiol*. 2017;65:563–76. doi: 10.23736/S0026-4725.17.04451-6
- 42 Arzilli F, Taddei S, Graziadei L, *et al.* Potassium-rich and sodium-poor salt reduces blood pressure in hospitalized patients. *J Hypertens Suppl Off J Int Soc Hypertens*. 1986;4:S347-350.

- 43 Baek SH, Ahn JW, Lee HR, *et al.* Anti-hypertensive effect of a solar salt diet in elderly hypertensive patients: A preliminary randomized, double-blind clinical trial. *Korean J Health Promot.* 2016;15:98–107.
- 44 Barros CL de A, Sousa ALL, Chinem BM, *et al.* Impact of light salt substitution for regular salt on blood pressure of hypertensive patients. *Arq Bras Cardiol.* 2015;104:128–35. doi: 10.5935/abc.20140174
- 45 Bernabe-Ortiz A, Sal Y Rosas VG, Ponce-Lucero V, *et al.* Effect of salt substitution on community-wide blood pressure and hypertension incidence. *Nat Med.* 2020;26:374–8. doi: 10.1038/s41591-020-0754-2
- 46 Chang H-Y, Hu Y-W, Yue C-SJ, *et al.* Effect of potassium-enriched salt on cardiovascular mortality and medical expenses of elderly men. *Am J Clin Nutr.* 2006;83:1289–96. doi: 10.1093/ajcn/83.6.1289
- 47 Che L, Song W, Zhang Y, *et al.* A randomized, double-blind clinical trial to evaluate the blood pressure lowering effect of low-sodium salt substitution on middle-aged and elderly hypertensive patients with different plasma renin concentrations. *J Clin Hypertens.* 2022;24:140–7. doi: 10.1111/jch.14396
- 48 China Salt Substitute Study Collaborative Group. Salt substitution: A low-cost strategy for blood pressure control among rural Chinese. A randomized, controlled trial. *J Hypertens.* 2007;25:2011–8. doi: 10.1097/HJH.0b013e3282b9714b
- 49 Crouch SH, Ware LJ, Norris SA, *et al.* Comparing a range of potassium-enriched low sodium salt substitutes to common salt: Results of taste and visual tests in south african adults. *Nutr Metab Cardiovasc Dis.* 2024;34:903–10. doi: 10.1016/j.numecd.2023.12.015
- 50 Geleijnse JM, Witterman JCM, Bak AAA, *et al.* Reduction in blood pressure with a low sodium, high potassium, high magnesium salt in older subjects with mild to moderate hypertension. *BMJ.* 1994;309:436–40. doi: 10.1136/bmj.309.6952.436
- 51 Gilleran G, O’Leary M, Bartlett WA, *et al.* Effects of dietary sodium substitution with potassium and magnesium in hypertensive type II diabetics: A randomised blind controlled parallel study. *J Hum Hypertens.* 1996;10:517–21.
- 52 Hu J, Zhao L, Thompson B, *et al.* Effects of salt substitute on home blood pressure differs according to age and degree of blood pressure in hypertensive patients and their families. *Clin Exp Hypertens.* 2018;40:664–72. doi: 10.1080/10641963.2018.1425415
- 53 Kawasaki T, Itoh K, Kawasaki M. Reduction in blood pressure with a sodium-reduced, potassium- and magnesium-enriched mineral salt in subjects with mild essential hypertension. *Hypertens Res.* 1998;21:235–43. doi: 10.1291/hypres.21.235
- 54 Li N, Prescott J, Wu Y, *et al.* The effects of a reduced-sodium, high-potassium salt substitute on food taste and acceptability in rural northern China. *Br J Nutr.* 2008;101:1088–93. doi:

10.1017/S0007114508042360

- 55 Li N, Yan LL, Niu W, *et al.* The effects of a community-based sodium reduction program in rural China – a cluster-randomized trial. *PLOS ONE*. 2016;11:e0166620. doi: 10.1371/journal.pone.0166620
- 56 Li Z, Yang H, Wu Y, *et al.* Relationship of the dietary sodium substitution with low sodium and high potassium salt with the blood pressure among middle aged and elderly people: A randomized controlled study. *Chin J Geriatr*. Published Online First: 2014.
- 57 Maleki A, Soltanian AR, Zeraati F, *et al.* The flavor and acceptability of six different potassium-enriched (sodium reduced) iodized salts: A single-blind, randomized, crossover design. *Clin Hypertens*. 2016;22:18. doi: 10.1186/s40885-016-0054-9
- 58 Mickelsen O, Makdani D, Gill J, *et al.* Sodium and potassium intakes and excretions of normal men consuming sodium chloride or a 1:1 mixture of sodium and potassium chlorides. *Am J Clin Nutr*. 1977;30:2033–40. doi: 10.1093/ajcn/30.12.2033
- 59 Mu J, Liu Z, Liu F, *et al.* Family-based randomized trial to detect effects on blood pressure of a salt substitute containing potassium and calcium in hypertensive adolescents. *Am J Hypertens*. 2009;22:943–7. doi: 10.1038/ajh.2009.136
- 60 Mu J, Liu Z, Yang J, *et al.* [long term observation in effects of potassium and calcium supplementation on arterial blood pressure and sodium metabolism in adolescents with higher blood pressure]. *Zhonghua Yu Fang Yi Xue Za Zhi*. 2003;37:90–2.
- 61 Neal B, Wu Y, Feng X, *et al.* Effect of salt substitution on cardiovascular events and death. *N Engl J Med*. 2021;385:1067–77. doi: 10.1056/NEJMoa2105675
- 62 Omvik P, Myking OL. Unchanged central hemodynamics after six months of moderate sodium restriction with or without potassium supplement in essential hypertension. *Blood Press*. 1995;4:32–41. doi: 10.3109/08037059509077565
- 63 Pan W-H, Lai Y-H, Yeh W-T, *et al.* Intake of potassium- and magnesium-enriched salt improves functional outcome after stroke: A randomized, multicenter, double-blind controlled trial. *Am J Clin Nutr*. 2017;106:1267–73. doi: 10.3945/ajcn.116.148536
- 64 Pereira MADG, Galvão R, Zanella MT. Efeitos da suplementação de potássio via sal de cozinha sobre a pressão arterial e a resistência à insulina em pacientes obesos hipertensos em uso de diuréticos. *Rev Nutr*. 2005;18:5–17. doi: 10.1590/S1415-52732005000100001
- 65 Sarkkinen ES, Kastarinen MJ, Niskanen TH, *et al.* Feasibility and antihypertensive effect of replacing regular salt with mineral salt -rich in magnesium and potassium- in subjects with mildly elevated blood pressure. *Nutr J*. 2011;10:88. doi: 10.1186/1475-2891-10-88
- 66 Suppa G, Pollavini G, Alberti D, *et al.* Effects of a low-sodium high-potassium salt in hypertensive patients treated with metoprolol: A multicentre study. *J Hypertens*.

- 1988;6:787-790. doi: 10.1097/00004872-198810000-00004
- 67 Yang G-H, Zhou X, Ji W-J, *et al.* Effects of a low salt diet on isolated systolic hypertension: A community-based population study. *Medicine (Baltimore)*. 2018;97:e0342. doi: 10.1097/MD.00000000000010342
 - 68 Yu J, Thout SR, Li Q, *et al.* Effects of a reduced-sodium added-potassium salt substitute on blood pressure in rural Indian hypertensive patients: A randomized, double-blind, controlled trial. *Am J Clin Nutr*. 2021;114:185–93. doi: 10.1093/ajcn/nqab054
 - 69 Yuan Y, Jin A, Neal B, *et al.* Salt substitution and salt-supply restriction for lowering blood pressure in elderly care facilities: A cluster-randomized trial. *Nat Med*. 2023;29:973–81. doi: 10.1038/s41591-023-02286-8
 - 70 Zhang H, Li DF, Li WW, *et al.* Pp.02.34: Safety of long-term enriched potassium salt consumption and effect on blood pressure in Chinese. *J Hypertens*. 2015;33:e148–9. doi: 10.1097/01.hjh.0000467774.65584.f4
 - 71 Zhao X, Yin X, Li X, *et al.* Using a low-sodium, high-potassium salt substitute to reduce blood pressure among tibetans with high blood pressure: A patient-blinded randomized controlled trial. *PloS One*. 2014;9:e110131. doi: 10.1371/journal.pone.0110131
 - 72 Zhou B, Wang H-L, Wang W-L, *et al.* Long-term effects of salt substitution on blood pressure in a rural north Chinese population. *J Hum Hypertens*. 2013;27:427–33. doi: 10.1038/jhh.2012.63
 - 73 Zhou X, Liu J-X, Shi R, *et al.* Compound ion salt, a novel low-sodium salt substitute: From animal study to community-based population trial. *Am J Hypertens*. 2009;22:934–42. doi: 10.1038/ajh.2009.135
 - 74 Huang L, Li Q, Wu JH, *et al.* The contribution of sodium reduction and potassium increase to the blood pressure lowering observed in the salt substitute and stroke study. *J Hum Hypertens*. 2024;38:298–306. doi: 10.1038/s41371-024-00896-4
 - 75 Greer RC, Marklund M, Anderson CAM, *et al.* Potassium-enriched salt substitutes as a means to lower blood pressure: Benefits and risks. *Hypertension*. 2020;75:266–74. doi: 10.1161/HYPERTENSIONAHA.119.13241
 - 76 Tsai Y-C, Tsao Y-P, Huang C-J, *et al.* Effectiveness of salt substitute on cardiovascular outcomes: A systematic review and meta-analysis. *J Clin Hypertens Greenwich Conn*. 2022;24:1147–60. doi: 10.1111/jch.14562
 - 77 Brand A, Visser ME, Schoonees A, *et al.* Replacing salt with low-sodium salt substitutes (LSSS) for cardiovascular health in adults, children and pregnant women. *Cochrane Database Syst Rev*. 2022;8:CD015207. doi: 10.1002/14651858.CD015207
 - 78 Greenwood H, Barnes K, Clark J, *et al.* Long-Term Effect of Salt Substitution for

- Cardiovascular Outcomes : A Systematic Review and Meta-analysis. *Ann Intern Med.* 2024;177:643–55. doi: 10.7326/M23-2626
- 79 Chan A, Chan SW, Khanam M, *et al.* Factors affecting reductions in dietary salt consumption in people of Chinese descent: An integrative review. *J Adv Nurs.* 2022;78:1919–37. doi: 10.1111/jan.15237
 - 80 Bu T, Tang D, Liu Y, *et al.* Trends in dietary patterns and diet-related behaviors in China. *Am J Health Behav.* 2021;45:371–83. doi: 10.5993/AJHB.45.2.15
 - 81 Xiang H, Tao X, Guan X, *et al.* Contemporary Chinese dietary pattern: Where are the hidden risks? *Front Nutr.* 2022;9:997773. doi: 10.3389/fnut.2022.997773
 - 82 Aaron KJ, Sanders PW. Role of dietary salt and potassium intake in cardiovascular health and disease: A review of the evidence. *Mayo Clin Proc.* 2013;88:987–95. doi: 10.1016/j.mayocp.2013.06.005
 - 83 Richardson SI, Freedman BI, Ellison DH, *et al.* Salt sensitivity: A review with a focus on non-hispanic blacks and hispanics. *J Am Soc Hypertens.* 2013;7:170–9. doi: 10.1016/j.jash.2013.01.003
 - 84 Rayner BL, Spence JD. Hypertension in blacks: Insights from africa. *J Hypertens.* 2017;35:234–9. doi: 10.1097/HJH.0000000000001171
 - 85 Kovesdy CP, Matsushita K, Sang Y, *et al.* Serum potassium and adverse outcomes across the range of kidney function: A CKD prognosis consortium meta-analysis. *Eur Heart J.* 2018;39:1535–42. doi: 10.1093/eurheartj/ehy100
 - 86 Marklund M, Singh G, Greer R, *et al.* Estimated population wide benefits and risks in China of lowering sodium through potassium enriched salt substitution: Modelling study. *BMJ.* 2020;m824. doi: 10.1136/bmj.m824
 - 87 Zhang J, Du W, Huang F, *et al.* Longitudinal study of dietary patterns and hypertension in adults: China health and nutrition survey 1991–2018. *Hypertens Res.* 2023;46:2264–71. doi: 10.1038/s41440-023-01322-x
 - 88 Kovesdy CP. Management of hyperkalaemia in chronic kidney disease. *Nat Rev Nephrol.* 2014;10:653–62. doi: 10.1038/nrneph.2014.168
 - 89 Ab Rahman R, Awang H, Syed Sulaiman SA. Hyperkalemia in patients with type 2 diabetes mellitus: Risk factors and clinical outcomes. *Eur J Clin Med.* 2023;4:18–24. doi: 10.24018/clinicmed.2023.4.4.291
 - 90 Goia-Nishide K, Coregliano-Ring L, Rangel ÉB. Hyperkalemia in diabetes mellitus setting. *Diseases.* 2022;10:20. doi: 10.3390/diseases10020020
 - 91 Gupta N, Prasad P. Hyperkalemia in diabetes: Newer insights into mechanism and treatment.

- Postgrad Med J.* 2022;98:e45–e45. doi: 10.1136/postgradmedj-2021-140226
- 92 Dépret F, Peacock WF, Liu KD, *et al.* Management of hyperkalemia in the acutely ill patient. *Ann Intensive Care.* 2019;9:32. doi: 10.1186/s13613-019-0509-8
- 93 Mclean A, Nath M, Sawhney S. Population epidemiology of hyperkalemia: Cardiac and kidney long-term health outcomes. *Am J Kidney Dis.* 2022;79:527-538.e1. doi: 10.1053/j.ajkd.2021.07.008
- 94 Liem DG, Miremadi F, Zandstra EH, *et al.* Health labelling can influence taste perception and use of table salt for reduced-sodium products. *Public Health Nutr.* 2012;15:2340–7. doi: 10.1017/S136898001200064X
- 95 Sinopoli DA, Lawless HT. Taste properties of potassium chloride alone and in mixtures with sodium chloride using a check-all-that-apply method. *J Food Sci.* 2012;77. doi: 10.1111/j.1750-3841.2012.02862.x
- 96 Liu Y, Chu H, Peng K, *et al.* Factors associated with the use of a salt substitute in rural China. *JAMA Netw Open.* 2021;4:e2137745. doi: 10.1001/jamanetworkopen.2021.37745
- 97 Sodium reduction. <https://www.who.int/news-room/fact-sheets/detail/sodium-reduction> (accessed 13 October 2025)

Table 1 Classification and definitions of salt substitutes

Classification Strategy	Category	Definition and Examples
Strategy 1: By Primary Composition (components $\geq 3\%$ of salt mixture)	K-salt	KCl + NaCl mixtures (e.g., 75% NaCl + 25% KCl)
	K-Ca-salt	KCl + NaCl + Calcium compounds (e.g., 65% NaCl + 30% KCl + 5% calcium salts)
	K-Mg-salt	KCl + NaCl + Magnesium compounds (e.g., 42.85% NaCl + 42.85% KCl + 14.3% MgSO ₄)
	K-Ca-Mg-salt	KCl + NaCl + Calcium + Magnesium compounds (e.g., 41% NaCl + 41% KCl + 17% Mg salts + 1% other minerals)
	Chitosan salt	NaCl + Chitosan (polysaccharide derived from chitin) (e.g., 97% NaCl + 3% chitosan)
	<i>Hypothesis</i>	<i>Variations in mineral composition may lead to different health effects due to distinct physiological mechanisms of different components.</i>
Strategy 2: By Potassium Content	High-potassium	50-65% KCl (e.g., 50% NaCl + 50% KCl)
	Moderate-potassium	25-40% KCl (e.g., 75% NaCl + 25% KCl)
	Low-potassium	5-20% KCl (e.g., 85% NaCl + 15% KCl)
	<i>Hypothesis</i>	<i>Potassium levels affect health outcomes through potassium's independent effects on blood pressure regulation and cardiovascular function.</i>
Strategy 3: By Sodium Content	Reduced-sodium	80-97% NaCl (e.g., 85% NaCl + 15% KCl)
	Low-sodium	60-79% NaCl (e.g., 75% NaCl + 25% KCl)
	Very low-sodium	40-59% NaCl (e.g., 50% NaCl + 50% KCl)
	Extremely low-sodium	2-39% NaCl (e.g., 34% NaCl + 66% KCl)
	<i>Hypothesis</i>	<i>Sodium levels affect health outcomes through sodium restriction's effects on</i>

The same formulation may be classified differently under each approach. For example, 75% NaCl + 25% KCl would be classified as K-salt (Strategy 1), Moderate-potassium (Strategy 2), and Low-sodium (Strategy 3).

KCl: potassium chloride; NaCl: sodium chloride; MgSO₄: magnesium sulfate; K: potassium; Ca: calcium; Mg: magnesium.

Table 2 List of included studies and baseline characteristics

Study, year	Country	Study design	Duration of intervention	Description of participants	Sample size of IG (n)	Sample size of CG (n)	Mean age $\pm$ SD (years)	Mean BMI $\pm$ SD (years)	Mean SBP $\pm$ SD (years)	Mean DBP $\pm$ SD (years)	Intervention name (as reported)	Components of intervention	Control name (as reported)	Components of control	Outcomes
Allaert, 2013	France	Crossover	8 weeks	Patients with stage 1 hypertension (SBP 140-159 mmHg or DBP 90-99 mmHg)	22	19	58.59 $\pm$ 11.98	25.10 $\pm$ 3.80	149.43 $\pm$ 4.71	93.35 $\pm$ 3.15	Chitosan salt	97% NaCl and 3% chitosan	Sea salt	100% NaCl	DBP SBP Any AE related to intervention Withdrawal due to any AE Withdrawal due to any reason

Allaert, 2017	France	Parallel	8 weeks	Patients with pre-hypertension (SBP 130-139 mmHg with or without DBP 80-89 mmHg)	22	19	51.00 ± 16.00	NR	133.78 ± 4.17	82.29 ± 6.00	Chitosan salt	97% NaCl and 3% chitosan	Sea salt	100% NaCl	DBP SBP Serum potassium Serum sodium Urinary potassium Urinary sodium Any AE related to intervention Non-fatal cardiovascular
---------------	--------	----------	---------	--	----	----	---------------	----	---------------	--------------	---------------	--------------------------	----------	-----------	--

															ular event s Acce ptanc e Withd rawal s due to any AE Withd rawal s due to any reaso n
Arzilli, 1986	Italy	Parallel	8 days	Patients with mild hypertension (supine DBP > 95 mmHg after 5 days of hospita	5	5	Range: 28 to 53	NR	NR	NR	K-rich/ Na-poor or salt	NR	Common salt	NR	DBP SBP Withdrawal s due to any AE

				lization)											
Baek, 2015	Korea	Parallel	8 weeks	Patients with hypertension (SBP ≥ 140 mmHg, DBP ≥ 90 mmHg, or used anti-hypertensive medication)	18	20	79.76 $\pm$ 7.52	24.40 $\pm$ 4.37	128.29 $\pm$ 12.1	75.05 $\pm$ 15.41	Solar salt	85.03% NaCl and other trace minerals	Refined salt	99.05% NaCl and other trace minerals	DBP SBP Serum potassium Serum sodium Urinary potassium Urinary sodium Withdrawals due to any reason
Barros, 2014	Brazil	Parallel	4 weeks	Patients with uncontrolled	19	16	55.50 $\pm$ 7.40	29.38 $\pm$ 5.55	143.17 $\pm$ 14.2	90.43 $\pm$ 8.99	Light salt	130 mg of sodium	Regulated salt	390 mg of sodium	DBP SBP Urinary

				rolled hypertension (SBP ≥140 mmHg or DBP ≥90 mmHg)				6			m, 346 mg of potassium and 44 mcg of iodine per gram		m and 25 mcg of iodine per gram	ry potassium Urinary sodium Any AE related to intervention Withdrawals due to any reason
Bernabe-Ortiz, 2020	Peru	Stepped-wedge cluster	15 months	General individuals	2376	43.30 ± 17.20	NR	113. 10 ± 17.0 0	72.0 0 ± 10.1 0	Salt substitute	75% NaCl and 25% KCl	Common salt	100% NaCl	DBP SBP Urinary potassium Urinary sodium

															m Any AE relate d to interv entio n Non-f atal cardi ovasc ular event s
Chan g, 2006	Taiw an, Chin a	Clust er	2.6 years	Genera l individ uals	768	1213	74.86 ± 6.86	23.30 ± 3.50	130. 93 ± 20.1 3	71.3 2 ± 10.8 0	Pota ssiu m-en riche d salt	49% NaCl, 49% KCl, and 2% other additi ves	Regul ar salt	99.6 % NaCl and 0.4% other additi ves	All-c ause morta lity Cardi ovasc ular morta lity
Che, 2022	Chin a	Parall el	12 months	Patient s with primar y hypert	162	160	62.57 ± 4.61	26.16 ± 3.23	134. 28 ± 13.4 2	77.8 0 ± 10.6 4	Low -sodi um salt	43% NaCl, 32% KCl, and	Norm al salt	100% NaCl	DBP SBP Seru m potas

				ension (SBP ≥140 mmHg , DBP ≥90 mmHg , or used anti-hy pertens ive medica tion)							25% other additi ves			sium Seru m sodiu m Urina ry potas sium Urina ry sodiu m Withd rawal s due to any reaso n
Crouc h, 2024	Sout h Afri ca	Cross over	NR	Genera l individ uals	56	21.00 ± 3.33	NR	NR	NR	Salt subst itute	65% NaCl and 35% KCl; 50% NaCl and 50% KCl;	Usual table salt	100% NaCl	Acce ptanc e

												34% NaCl and 66% KCl; 100% KCl			
CSSS , 2007	China	Parallel	12 months	Individuals at risk of vascular disease (SBP ≥160 mmHg, or age ≥55 years)	306	302	59.99 ± 9.89	26.00 ± 3.60	159.00 ± 25.48	93.00 ± 13.99	Salt substitute	65% NaCl, 25% KCl, and 10% MgSO4	Normal salt	100% NaCl	DBP SBP Urinary potassium Urinary sodium Any AE related to intervention Non-fatal cardiovascular

															event s All-c ause morta lity Withd rawal s due to any reaso n
Geleij nse, 1994	Neth erlan ds	Parall el	24 weeks	Patient s with hypert ension (SBP 140-20 0 mmHg or DBP 85-110 mmHg)	49	51	66.41 ± 4.58	27.10 ± 3.40	157. 75 ± 13.8 5	90.3 1 ± 9.22	Min eral salt	41% NaCl, 41% KCl, 17% Mg salts, and 1% other trace miner als	Com mon salt	100% NaCl	DBP SBP Seru m potas sium Seru m sodiu m Urina ry potas sium Urina ry sodiu

															m Withd rawal s due to any reaso n
Giller an, 1996	UK	Parall el	9 months	Patient s with hypert ension (SBP > 160 mmHg , DBP > 95 mmHg , or used anti-hy pertens ive medica tion) and type 2 diabete s	20	20	62.50 ± 7.80	28.10 ± 4.60	163. 20 ± NR	91.2 0 ± NR	Salt subst itute	50% NaCl, 40% KCl, and 10% MgS O4	Regul ar salt	NR	DBP SBP Urina ry potas sium Urina ry sodiu m Non-f atal cardi ovasc ular event s Withd rawal s due to any reaso

															n
Hu, 2018	China	Cluster	12 months	Patients with mild to moderate hypertension (SBP ≥140 mmHg or DBP ≥90 mmHg), and their adult family members	301	299	49.91 ± 16.26	25.89 ± 3.85	42 ± 17.18	78.41 ± 9.93	Salt substitute	65% NaCl, 25% KCl, and 10% MgSO4	Regular salt	100% NaCl	DBP SBP Urinary potassium Urinary sodium Any AE related to intervention Acceptance Withdrawals due to any AE Withdrawals due

															to any reason
Kawa saki, 1998	Japa n	Parall el	5 weeks	Partici pants with normal BP (SBP <140 mmHg and DBP <90 mmHg) and patient s with hypert ension (SBP > 140 mmHg or DBP > 90 mmHg)	21	20	65.85 ± 7.40	22.90 ± 2.70	135. 10 ± 17.6 3	76.8 9 ± 10.7 2	Min eral salt	229 mg of sodiu m, 101.1 mg of potass ium and 12.4 mg of magn esium per gram	Regul ar salt	390 mg of sodiu m, 1.3 mg of potass ium and 0.21 mg of magn esium per gram	DBP SBP Seru m potas sium Seru m sodiu m Urina ry potas sium Urina ry sodiu m Withd rawal s due to any reason
Li,	Chin	Parall	12	Individ	306	302	59.99	26.00	159.	93.0	Salt	65%	Norm	100%	Acce

2009	a	el	months	uals at risk of CVD			± 9.89	± 3.60	00 ± 25.48	0 ± 24.49	subst itute	NaCl, 25% KCl, and 10% MgSO ₄	al salt	NaCl	ptanc e
Li, 2014	China	Cluster	2 months	General individual uals	253	263	62.15 ± 5.45	NR	127.83 ± 14.26	81.64 ± 10.94	Salt subst itute	70% NaCl and 30% KCl	No interv ention	NR	DBP SBP Any AE relate d to interv entio n Withd rawal s due to any AE
Li, 2016	China	Cluster	18 months	General individual uals	1294	1272	55.00 ± 14.51	24.00 ± 3.50	NR	NR	Salt subst itute	70% NaCl and 30% KCl	Regul ar salt	NR	DBP SBP Urina ry potas sium Urina ry

															sodium Withdrawals due to any reason
Male ki, 2016	Iran	Cross over	NR	General individuals	100	38.13 ± 10.70	NR	NR	NR	Salt substitute	95% NaCl and 5% KCl; 90% NaCl and 10% KCl; 85% NaCl and 15% KCl; 80% NaCl and 20% KCl; 75%	Regular salt	100% NaCl	Acceptance Withdrawals due to any AE	

												NaCl and 25% KCl; 70% NaCl and 30% KCl			
Mickelsen, 1977	USA	Parallel	1 months	Men with normal BP	10	10	27.40 ± 6.96	NR	119.80 ± 6.19	74.20 ± 7.65	Salt substitute	49% NaCl, 49.6% KCl, and 1.4% other additives	Regular salt	99.6% NaCl and 0.4% other additives	DBP SBP Serum potassium Serum sodium Urinary potassium Urinary sodium Withdrawal

															s due to any reason
Mu, 2003	China	Parallel	2 years	Young adults with high BP (SBP ≥ 90 th percentile by age and sex)	110	110	20.55 $\pm$ 2.10	21.90 $\pm$ 2.90	124.50 $\pm$ 11.98	76.50 $\pm$ 10.50	Add ed-K -and -Ca salt	Add 10 mmol of potassium and 10 mmol of calcium per person per day	Regular salt	NR	DBP SBP Urinary potassium Urinary sodium
Mu, 2009	China	Cluster	2 years	Family members of the adolescents	334	254	NR	NR	NR	NR	Add ed-K -and -Ca salt	Add 10 mmol of potassium and 10 mmol of	No intervention	NR	DBP SBP Urinary potassium Urinary sodium

												calciu m per perso n per day			
Neal, 2021	Chin a	Clust er	4.74 years	Genera l individ uals	10504	10491	65.35 ± 8.50	24.80 ± 3.60	154. 00 ± 23.5 0	89.2 5 ± 14.0 0	Salt subst itute	75% NaCl and 25% KCl	Regul ar salt	100% NaCl	DBP SBP Urina ry potas sium Urina ry sodiu m Non-f atal cardi ovasc ular event s All-c ause morta lity Cardi ovasc ular

															morta lity Withd rawal s due to any reaso n
Omvi k, 1995	Nor way	Parall el	6 months	Patient s with mild to moder ate hypert ension (SBP ≥140 mmHg or DBP ≥90 mmHg)	20	20	44.30 ± NR	NR	151. 50 ± NR	90.0 0 ± NR	Pans alt	57% NaCl, 28% KCl, 12% MgS O4, and 2% lysine	Regul ar salt	100% NaCl	DBP SBP Seru m potas sium Seru m sodiu m Urina ry potas sium Urina ry sodiu m Non-f atal cardi

														ovascular event s Withdrawals due to any reason
Pan, 2017	Taiwan, China	Parallel	6 months	Patients after stroke	192	99	64.64 ± 9.96	NR	137.85 ± 18.72	82.30 ± 10.82	IG1: K salt; IG 2: K/Mg salt IG 1: 50% NaCl; 50% KCl; IG 2: NaCl, 42.85% KCl, and 14.3% MgSO ₄	Regular salt	NR	Any AE related to intervention Non-fatal cardiovascular event s All-cause mortality Withdraw

															rawal s due to any AE Withd rawal s due to any reaso n
Pereira, 2005	Brazil	Parallel	12 weeks	Obese patients with hypertension	12	10	47.54 ± 11.19	32.50 ± 13.20	137. 75 ± 9.09	93.1 2 ± 4.63	Salt substitute	50% NaCl and 50% KCl	Common salt	100% NaCl	DBP SBP Serum potassium Serum sodium Urinary potassium Urinary sodium Withd

															rawal s due to any reaso n
Sarkk inen, 2011	Finla nd	Parall el	8 weeks	Patient s with mild hypert ension (SBP 130-15 9 mmHg or DBP 85-99 mmHg)	22	23	55.47 ± 11.47	28.00 ± 3.00	139. 47 ± 8.56	87.5 1 ± 6.48	Sma rt salt	50% NaCl, 25% KCl, and 25% magn esium ammo nium potass ium chlori de, hydra te	Regul ar salt	100% NaCl	DBP SBP Urina ry potas sium Urina ry sodiu m Non-f atal cardi ovasc ular event s Withd rawal s due to any reaso n
Suppa	Italy	Parall	4 weeks	Patient	163	159	47.45	NR	NR	NR	Diet	50%	Com	100%	DBP

, 1988		el		s with mild to moderate hypertension (supine DBP ≥ 95 mmHg)			$\pm$ NR				ary salt	NaCl, 25% KCl, and 15% potassium citrate	mon salt	NaCl	SBP Urinary potassium Urinary sodium Withdrawals due to any reason
Yang, 2018	China	Parallel	6 months	Patients with mild to moderate hypertension (SBP ≥ 140 mmHg , DBP ≥ 90 mmHg , or	62	64	66.58 $\pm$ 5.98	25.20 $\pm$ 3.80	158.48 $\pm$ 12.15	82.67 $\pm$ 6.97	Low sodium salt	65% NaCl, 30% KCl, 4% calcium citrate tetrahydrate , and 1% calcium	Normal salt	NR	DBP SBP Urinary potassium Urinary sodium Withdrawals due to any

				used anti-hy pertens ive medica tion)								carbo nate			AE Withd rawal s due to any reaso n
Yu, 2021	India	Parall el	3 months	Patient s with hypert ension diagno sed by a health profess ional	252	250	61.60 ± 12.02	23.10 ± 4.70	132. 45 ± 21.4 1	83.3 0 ± 12.5 5	Salt subst itute	70% NaCl and 30% KCl	Regul ar salt	100% NaCl	DBP SBP Urina ry potas sium Urina ry sodiu m Any AE relate d to interv entio n Acce ptanc e Withd rawal

															s due to any AE Withdrawals due to any reason
Yuan, 2023	China	Cluster	24 months	Elderly individuals in residential care facilities	406	363	70.47 ± 9.15	NR	137.99 ± 21.80	81.20 ± 12.05	Salt substitute	62.5% NaCl, 25% KCl, and 12.5% dried food ingredient flavorings	Usual salt	100% NaCl	DBP SBP Urinary potassium Urinary sodium m Non-fatal cardiovascular event s All-cause mortality

															lity Withd rawal s due to any AE Cardi ovasc ular morta lity Withd rawal s due to any reaso n
Zhan g, 2015	Chin a	Clust er	14 months	Partici pants living in nursin g houses	1105	947	65.00 ± NR	NR	150. 00 ± 27.2 2	82.5 4 ± 12.7 9	Salt subst itute	50% NaCl and 50% KCl	Norm al salt	NR	DBP SBP Seru m potas sium Any AE relate d to interv entio

															n
Zhao, 2014	China	Parallel	3 months	Patients with hypertension (SBP ≥ 140 mmHg)	141	141	63.15 $\pm$ 11.19	23.70 $\pm$ 3.10	85 $\pm$ 22.83	176. 104. 50 $\pm$ 12.61	Salt substitute	65% NaCl, 25% KCl, and 10% MgSO ₄	Regular salt	100% NaCl	DBP SBP All-cause mortality Withdrawals due to any reason
Zhou, 2009	China	Parallel	6 months	Patients with mild to moderate hypertension (SBP ≥ 140 mmHg or DBP ≥ 90 mmHg) or normal	119	129	66.63 $\pm$ 6.15	24.58 $\pm$ 3.40	141. 80 $\pm$ 19.89	78.7 7 $\pm$ 9.09	Compound ion salt	65% NaCl, 30% KCl, and 5% calcium salts	Normal salt	NR	DBP SBP Urinary potassium Urinary sodium Non-fatal cardiovascular event

				BP (SBP <140 mmHg or DBP <90 mmHg)											s Cardi ovasc ular morta lity Withd rawal s due to any reaso n
Zhou, 2013	China	Cluster	2 years	Patient s with hypert ension (SBP > 140 mmHg or DBP > 90 mmHg) and their family membe rs had an	224	238	46.52 ± 1.12	26.00 ± 0.00	151. 94 ± 2.83	89.9 7 ± 1.41	Salt subst itute	65% NaCl, 25% KCl and 10% MgSO ₄	Normal salt	100% NaCl	DBP SBP Urinary potas sium Urinary sodium All-cause mortality Withdrawal s due

				estimat ed daily sodium intake of ≥260m mol per day												to any reaso n
--	--	--	--	---	--	--	--	--	--	--	--	--	--	--	--	----------------------

SBP: systolic blood pressure; DBP: diastolic blood pressure; IG: intervention group; BMI: body mass index; NR: not report; NaCl: sodium chloride; KCl: potassium chloride; MgSO4: magnesium sulfate.

Table 3 Summary of network meta-analysis results

Superior (for serum and urinary electrolytes: important decrease)
relative to regular salt with moderate to high certainty

Trivial or no benefit (for serum and urinary electrolytes: trivial or
no decrease) relative to regular salt with moderate to high certainty

Superior (for serum and urinary electrolytes: important decrease)
relative to regular salt with very low to low certainty

Trivial or no benefit (for serum and urinary electrolytes: trivial or
no decrease) relative to regular salt with very low to low certainty

Table 3a Summary of network meta-analysis results of primary composition of salt substitutes

Outcomes	Base line risk	MID	K-salt		K-Ca-salt		K-Mg-salt		K-Ca-Mg-salt		Chitosan salt	
			Studies (participants) included in analysis, n	Estimate (95% CI)	Studies (participants) included in analysis, n	Estimate (95% CI)	Studies (participants) included in analysis, n	Estimate (95% CI)	Studies (participants) included in analysis, n	Estimate (95% CI)	Studies (participants) included in analysis, n	Estimate (95% CI)
All-cause mortality	Intermediate: 62 per 1000 persons	5 per 1000 persons	4 (24, 538)	-6 (-10 to -2)	-	-	4 (1,499)	11 (-27 to 87)	-	-	-	-
	High: 143 per 1000 persons	5 per 1000 persons		-14 (-23 to -6)				24 (-61 to 202)				
Cardiovascular mortality	Intermediate: 29 per	5 per 1000 per	3 (24, 384)	-6 (-10 to -1)	-	-	-	-	-	-	-	--

	1000 persons	sons										
	High : 92 per 1000 persons	5 per 1000 persons		-19 (-31 to -4)								
Non-fatal cardiovascular events	Intermediate: 75 per 1000 persons	10 per 1000 persons	4 (25, 511)	-23 (-38 to -3)	-	-	5 (981)	22 (-36 to 166)	-	-	1 (41)	-53 (-74 to 461)
	High : 250 per 1000 persons	10 per 1000 persons		-78 (-125 to -10)				73 (-120 to 553)				-178 (-248 to 1535)
SBP	-	3 mm Hg	12 (23, 993)	-4.00 (-5.67 to -2.32)	4 (1,362)	-4.09 (-6.96 to -1.22)	9 (2,148)	-6.43 (-8.62 to -4.25)	1 (38)	-17.50 (-27.36 to -7.64)	2 (80)	-10.87 (-15.67 to -6.07)
DBP	-	2 mm Hg	12 (23, 993)	-1.42 (-2.57 to -0.26)	4 (1,362)	-2.79 (-4.80 to -0.78)	9 (2,148)	-2.75 (-4.32 to -1.19)	1 (38)	-9.67 (-16.80 to -2.53)	2 (80)	-7.43 (-11.62 to -3.24)
AEs related to	20 per 1000	30 per 100	2 (2,087)	2 (-9 to 20)	-	-	2 (1,208)	7 (-11 to 38)	1 (41)	-25 (-30 to 72)	-	-

inter venti on	pers ons	0 per son s										
With draw als due to any reaso n	180 per 1000 pers ons	30 per 100 0 per son s	8 (22, 934)	-18 (-27 to -9)	1 (248)	10 3 (-5 6 to 45 5)	9 (2,3 24)	49 (9 to 97)	1 (43)	-54 (-157 to 499)	1 (40)	-18 (-1 69 to 22 41)
Seru m potas sium	-	0.1 4 mm ol/ L	2 (2,4 16)	0.11 (0.07 to 0.15)	-	-	3 (180)	0.17 (0.08 to 0.26)	1 (38)	0.27 (-0.06 to 0.60)	1 (39)	0.1 0 (-0 .06 to 0.2 6)
Seru m sodiu m	-	0.7 5 mm ol/ L	3 (364)	-0.15 (-1.30 to 0.99)	-	-	3 (180)	-0.48 (-1.52 to 0.57)	1 (38)	-0.50 (-3.86 to 2.86)	1 (39)	-0. 10 (-2 .22 to 2.0 2)
Urina ry potas sium	-	8.9 7 mm ol/2 4h	10 (7,6 19)	17.22 (12.20 to 22.24)	4 (1,3 62)	6.4 0 (-0 .67 to 13. 47)	8 (1,9 35)	12.26 (5.75 to 18.77)	1 (38)	20.75 (-0.10 to 41.60)	1 (38)	-2. 40 (-4 5.1 9 to 40. 39)
Urina ry sodiu m	-	15. 31 mm ol/2 4h	10 (7,6 19)	-19.33 (-36.0 5 to -2.62)	4 (1,3 62)	-33 .97 (-5 8.4 5 to -9. 48)	8 (1,9 35)	-5.90 (-24.86 to 13.07)	1 (38)	2.75 (-70.33 to 75.83)	1 (38)	11. 55 (-5 3.2 0 to 76. 30)

SBP: Systolic blood pressure; DBP: Diastolic blood pressure; K-salt: Salt substitute containing potassium chloride in varying proportions; K-Ca-salt: Combination of potassium chloride and calcium

carbonate; K-Mg-salt: Combination of potassium chloride and magnesium sulfate; K-Ca-Mg-salt: Combination of potassium chloride, calcium carbonate, and magnesium sulfate; Chitosan salt: Salt substitute containing chitosan as the primary component.

Table 3b Summary of network meta-analysis results of different potassium chloride contents of salt substitutes

Outcomes	Baseline risk	MID	High-potassium SS		Moderate-potassium SS		Low-potassium SS	
			Studies (participants included in analysis, n)	Estimate (95% CI)	Studies (participants included in analysis, n)	Estimate (95% CI)	Studies (participants included in analysis, n)	Estimate (95% CI)
All-cause mortality	Intermediate: 62 per 1000 persons	5 per 1000 persons	2 (1,931)	-2 (-11 to 7)	6 (24,106)	-7 (-11 to -3)	-	-
	High: 143 per 1000 persons	5 per 1000 persons		-6 (-24 to 17)		-16 (-24 to -7)		
Cardiovascular mortality	Intermediate: 29 per 1000 persons	5 per 1000 persons	1 (1,777)	-10 (-17 to 1)	2 (22,607)	-5 (-9 to 0)	-	-
	High: 92 per 1000 persons	5 per 1000 persons		-33 (-55 to 3)		-16 (-28 to -1)		-
Non-fatal cardiovascular events	Intermediate: 75 per 1000 persons	10 per 1000 persons	1 (291)	78 (-62 to 1677)	7 (25,945)	-20 (-35 to 0)	1 (41)	-53 (-74 to 461)

	s							
	High: 250 per 1000 person s	10 per 1000 pers ons		260 (-205 to 5590)		-68 (-115 to 0)		-178 (-248 to 1538)
SBP	-	3 mm Hg	4 (2,129)	-7.54 (-11.63 to -3.44)	18 (24,34 5)	-4.64 (-6.05 to -3.24)	6 (1,147)	-7.07 (-9.87 to -4.27)
DBP	-	2 mm Hg	4 (2,129)	-3.26 (-6.45 to -0.07)	18 (24,34 5)	-1.98 (-3.20 to -0.76)	6 (1,147)	-4.58 (-7.11 to -2.05)
AEs related to interve ntion	20 per 1000 person s	30 per 1000 pers ons	2 (2,087)	2 (-9 to 20)	2 (1,208)	7 (-11 to 38)	1 (41)	-25 (-30 to 72)
Withdr awals due to any reason	180 per 1000 person s	30 per 1000 pers ons	3 (208)	25 (-54 to 151)	15 (2529 8)	18 (-14 to 59)	2 (83)	-45 (-149 to 403)
Serum potassi um	-	0.14 mmo l/L	3 (2,094)	0.10 (0.05 to 0.14)	3 (461)	0.15 (0.09 to 0.20)	3 (118)	0.17 (0.05 to 0.29)
Serum sodium	-	0.75 mmo l/L	2 (42)	0.73 (-0.88 to 2.34)	3 (461)	-0.53 (-0.91 to -0.15)	3 (118)	-1.02 (-1.81 to -0.23)
Urinar y potassi um	-	8.97 mmo l/24h	3 (77)	31.04 (19.99 to 42.09)	16 (9,810)	12.70 (8.86 to 16.53)	5 (1,105)	7.05 (-0.53 to 14.62)
Urinar y sodium	-	15.3 1 mmo l/24h	3 (77)	-57.39 (-96.21 to -18.57)	16 (9,810)	-16.71 (-29.39 to -4.02)	5 (1,105)	1.99 (-23.30 to 27.28)

SS: Salt substitutes; SBP: Systolic blood pressure; DBP: Diastolic blood pressure; High-potassium SS: Salt substitutes with 50–65% potassium chloride; Moderate-potassium SS: Salt substitutes with 25–40% potassium chloride; dwdLow-potassium SS: Salt substitutes with 5–20% potassium chloride.

Table 3c Summary of network meta-analysis results of different sodium chloride contents of salt substitutes

Outcomes	Base line risk	MI D	Reduced-sodium SS		Low-sodium SS		Very Low-sodium SS		Extremely Low-sodium SS	
			Studies (participants) included in analysis, n	Estimate (95% CI)	Studies (participants) included in analysis, n	Estimate (95% CI)	Studies (participants) included in analysis, n	Estimate (95% CI)	Studies (participants) included in analysis, n	Estimate (95% CI)
All-cause mortality	Intermediate: 62 per 1000 persons	5 per 100 persons	-	-	5 (23,959)	-7 (-11 to -3)	2 (2,078)	-2 (-11 to 8)	-	-
	High: 143 per 1000 persons	5 per 100 persons				-17 (-26 to -7)		-4 (-24 to 19)		
Cardiovascular mortality	Intermediate: 29 per 1000 persons	5 per 100 persons	-	-	2 (22,607)	-5 (-9 to 0)	1 (1,777)	-10 (-17 to 1)	-	-
	High: 92 per 1000 persons	5 per 100 persons				-16 (-28 to -1)		-33 (-55 to 3)		
Non-fatal cardiovascular	Intermediate: 75	10 per 1000	1 (41)	-53 (-74 to 462)	4 (24,977)	-21 (-36 to 0)	4 (416)	14 (-48 to 220)	-	-

adverse events	per 1000 persons	persons								
	High : 250 per 1000 persons	10 per 100 persons								
SBP	-	3 mm Hg	5 (1,106)	-6.88 (-9.80 to -3.97)	13 (23,518)	-4.39 (-6.02 to -2.75)	9 (2,962)	-6.18 (-8.50 to -3.85)	1 (35)	-10.08 (-23.17 to 3.01)
DBP	-	2 mm Hg	5 (1,106)	-4.45 (-6.58 to -2.32)	13 (23,518)	-1.42 (-2.46 to -0.39)	9 (2,962)	-2.91 (-4.36 to -1.46)	1 (35)	-6.80 (-14.71 to 1.11)
AEs related to intervention	20 per 1000 persons	30 per 100 persons	1 (41)	-25 (-30 to 72)	2 (1,208)	7 (-11, 38)	1 (2,052)	2 (-10, 20)	1 (35)	47 (-27 to 1728)
Withdrawals due to any reason	180 per 1000 persons	30 per 100 persons	2 (83)	-45 (-149 to 403)	9 (24,586)	31 (-13 to 83)	8 (915)	4 (-43 to 68)	-	-
Serum potassium	-	0.14 mmol/L	2 (77)	0.13 (-0.01 to 0.28)	1 (41)	0.25 (0.04 to 0.46)	6 (2,555)	0.11 (0.08 to 0.15)	-	-
Serum sodium	-	0.75 mmol/L	2 (77)	-0.19 (-1.62 to 1.24)	1 (41)	-1.40 (-2.38 to -0.42)	5 (503)	-0.44 (-0.83 to -0.04)	-	-
Urinary potassium	-	8.97 mmol/24h	4 (1,064)	6.97 (-1.52 to 15.46)	11 (8,983)	11.27 (6.91 to 15.64)	8 (910)	24.54 (17.86 to 31.21)	1 (35)	-7.28 (-25.64 to 11.08)

Urinary sodium	-	15.31 mmol/24h	4 (1,064)	4.68 (-24.07 to 33.44)	11 (8,983)	-14.34 (-29.71 to 1.03)	8 (910)	-27.00 (-47.28 to -6.72)	1 (35)	-55.48 (-121.83 to 10.87)
-----------------------	---	----------------	-----------	------------------------	------------	-------------------------	---------	--------------------------	--------	---------------------------

SS: Salt substitutes; SBP: Systolic blood pressure; DBP: Diastolic blood pressure; Reduced-sodium SS: Salt substitutes with 80–97% sodium chloride; Low-sodium SS: Salt substitutes with 60–79% sodium chloride; Very low-sodium SS: Salt substitutes with 40–59% sodium chloride; Extremely low-sodium SS: Salt substitutes with 2–39% sodium chloride.

Certainty of evidence ratings were assessed for the populations actually studied in the included trials, without pre-specification of particular target populations. When applying this evidence to populations with substantially different dietary patterns, food preparation methods, baseline sodium intake, or genetic backgrounds compared to the study populations, practitioners may need to consider downgrading evidence certainty for indirectness.



